LLM Evaluation Report

SageMaker JumpStart models

Generated: 2026-09-28 20:07:19
Test Models: sagemaker/huggingface-od-nvidia-locateanything-3b
Judge Model: none (geometric IoU against exact ground truth; prompts: prompt_templates_grounding.json)
AWS Region: us-east-1
DeepEval: Disabled
Endpoints: huggingface-od-nvidia-locateanything-3b → jumpstart-dft-hf-od-nvidia-locatean-20260821-170355

Executive Summary

Model	Tests	Pass Rate	Avg Score	Avg Latency
sagemaker/huggingface-od-nvidia-locat...	13	92.3%	0.88	1712ms

Performance by Category

Category	Tests	Pass Rate	Avg Score	Status
ABSENT_OBJECT	3	100.0%	1.00	PASS
ATTRIBUTE_BINDING	3	66.7%	0.66	WARNING
LOCALIZATION_ACCURACY	2	100.0%	0.92	PASS
MULTI_INSTANCE	2	100.0%	0.93	PASS
SCALE_ROBUSTNESS	2	100.0%	0.97	PASS
SPATIAL_REASONING	1	100.0%	0.79	PASS

Model: sagemaker/huggingface-od-nvidia-locateanything-3b

Category: LOCALIZATION_ACCURACY

Test 1: Single Object Localization - PASS | Score: 0.98

Input Prompt:

```
Locate all the instances that matches the following description: red circle.
```

AI Response:

```
Model ref: red circle. the instances Distinct boxes returned: 1 (repeated 42x in raw output)
Expected boxes: 1 expected px=[220, 160, 420, 360] -> predicted px=[219, 159, 422, 363]
IoU=0.966 (match)
```

Method	Score	Status
Localization	0.97	PASS
Detection	1.00	PASS
Format	1.00	PASS

Observations: localization: 0.97 | detection: 1.00 | format: 1.00 | mean IoU 0.966 over 1 target(s) | output repeated 42x (token waste) | probe: One large centred object. Baseline: if this fails nothing else matters.

Recommendations: Accurate localization and count. No action needed.

Tokens: 0->0 | Latency: 2290ms | Time: 2026-09-28 20:06:59

Test 2: Off-Centre Object Localization - PASS | Score: 0.85

Input Prompt:

```
Locate all the instances that matches the following description: green square.
```

AI Response:

```
Model ref: green square. the instances that matches the following: Distinct boxes returned: 2
(repeated 1x in raw output) Expected boxes: 1 expected px=[60, 260, 200, 400] -> predicted
px=[60, 259, 201, 402] IoU=0.972 (match) extra prediction px=[0, 0, 639, 480] -> false positive
```

Method	Score	Status
Localization	0.97	PASS
Detection	0.67	WARNING
Format	1.00	PASS

Observations: localization: 0.97 | detection: 0.67 | format: 1.00 | mean IoU 0.972 over 1 target(s) | 1 false positive(s) | probe: Object away from the centre, to catch a centre-bias prior.

Recommendations: Count is wrong (missed targets or false positives). Review instance separation for this scene type.

Tokens: 0->0 | Latency: 651ms | Time: 2026-09-28 20:06:59

Category: MULTI_INSTANCE

Test 1: Two Instances With Distractor - PASS | Score: 0.98

Input Prompt:

```
Locate all the instances that matches the following description: red circle.
```

AI Response:

```
Model ref: red circle. Distinct boxes returned: 2 (repeated 21x in raw output) Expected boxes: 2
expected px=[50, 50, 150, 150] -> predicted px=[49, 49, 152, 152] IoU=0.943 (match) expected
px=[400, 300, 500, 400] -> predicted px=[399, 300, 501, 401] IoU=0.971 (match)
```

Method	Score	Status
Localization	0.96	PASS
Detection	1.00	PASS
Format	1.00	PASS

Observations: localization: 0.96 | detection: 1.00 | format: 1.00 | mean IoU 0.957 over 2 target(s) | output repeated 21x (token waste) | probe: Two targets plus a blue square distractor. Tests count and class purity.

Recommendations: Accurate localization and count. No action needed.

Tokens: 0->0 | Latency: 5581ms | Time: 2026-09-28 20:07:05

Test 2: Four Instance Counting - PASS | Score: 0.88

Input Prompt:

```
Locate all the instances that matches the following description: red circle.
```

AI Response:

```
Model ref: red circle. Distinct boxes returned: 6 (repeated 7x in raw output) Expected boxes: 4
expected px=[40, 200, 120, 280] -> predicted px=[40, 199, 122, 281] IoU=0.952 (match) expected
px=[180, 200, 260, 280] -> predicted px=[180, 200, 261, 281] IoU=0.975 (match) expected
px=[320, 200, 400, 280] -> predicted px=[319, 200, 401, 281] IoU=0.964 (match) expected
px=[460, 200, 540, 280] -> predicted px=[460, 200, 541, 281] IoU=0.975 (match) extra prediction
px=[540, 200, 640, 281] -> false positive extra prediction px=[640, 200, 640, 281] -> false
positive
```

Method	Score	Status
Localization	0.97	PASS
Detection	0.80	PASS
Format	0.83	PASS

Observations: localization: 0.97 | detection: 0.80 | format: 0.83 | mean IoU 0.967 over 4 target(s) | 2 false positive(s) | 1 malformed box(es) | output repeated 7x (token waste) | probe: Four identical targets in a row. Under/over-counting shows up here.

Recommendations: Usable with minor localization error.

Tokens: 0->0 | Latency: 3172ms | Time: 2026-09-28 20:07:08

Category: ATTRIBUTE_BINDING

Test 1: Colour Discrimination, Same Shape - PASS | Score: 0.98

Input Prompt:

```
Locate all the instances that matches the following description: blue circle.
```

AI Response:

```
Model ref: blue circle. the Distinct boxes returned: 1 (repeated 2x in raw output) Expected boxes: 1
expected px=[380, 140, 520, 280] -> predicted px=[378, 139, 522, 282] IoU=0.952 (match)
```

Method	Score	Status
Localization	0.95	PASS
Detection	1.00	PASS
Format	1.00	PASS

Observations: localization: 0.95 | detection: 1.00 | format: 1.00 | mean IoU 0.952 over 1 target(s) | probe: Two circles differing only in colour. Must bind colour to shape.

Recommendations: Accurate localization and count. No action needed.

Tokens: 0->0 | Latency: 644ms | Time: 2026-09-28 20:07:09

Test 2: Shape Discrimination, Same Colour - FAIL | Score: 0.15

Input Prompt:

```
Locate all the instances that matches the following description: red square.
```

AI Response:

```
Model ref: (none) Distinct boxes returned: 0 (repeated 0x in raw output) Expected boxes: 1
expected px=[380, 140, 520, 280] -> MISSED
```

Method	Score	Status
Localization	0.00	FAIL
Detection	0.00	FAIL
Format	1.00	PASS

Observations: localization: 0.00 | detection: 0.00 | format: 1.00 | mean IoU 0.000 over 1 target(s) | 1 missed | probe: Two red objects differing only in shape.

Recommendations: Count is wrong (missed targets or false positives). Review instance separation for this scene type.

Tokens: 0->0 | Latency: 332ms | Time: 2026-09-28 20:07:09

Test 3: Shape Discrimination, Rephrased Query - PASS | Score: 0.85

Input Prompt:

```
Locate all the instances that matches the following description: red rectangle.
```

AI Response:

```
Model ref: red rectangle. the Distinct boxes returned: 2 (repeated 21x in raw output) Expected
boxes: 1 expected px=[380, 140, 520, 280] -> predicted px=[380, 139, 522, 281] IoU=0.972
(match) extra prediction px=[401, 0, 640, 480] -> false positive
```

Method	Score	Status
Localization	0.97	PASS
Detection	0.67	WARNING
Format	1.00	PASS

Observations: localization: 0.97 | detection: 0.67 | format: 1.00 | mean IoU 0.972 over 1 target(s) | 1 false positive(s) | output repeated 21x (token waste) | probe: Identical scene to attr_shape_red with one word changed. Probes phrasing sensitivity: the endpoint returned 0 boxes for 'red square' but located the square for 'red rectangle' on 2026-09-28.

Recommendations: Count is wrong (missed targets or false positives). Review instance separation for this scene type.

Tokens: 0->0 | Latency: 2814ms | Time: 2026-09-28 20:07:12

Category: ABSENT_OBJECT

Test 1: Absent Object - Real Class - PASS | Score: 1.00

Input Prompt:

```
Locate all the instances that matches the following description: dog.
```

AI Response:

```
Model ref: dog. all the instances that matches the following description Distinct boxes returned: 0
(repeated 0x in raw output) Expected boxes: 0 Correctly returned no boxes.
```

Method	Score	Status
Localization	1.00	PASS
Detection	1.00	PASS
Format	1.00	PASS

Observations: localization: 1.00 | detection: 1.00 | format: 1.00 | absent-object case: clean | probe: Image has only a red circle. Returning any box is a hallucination.

Recommendations: Correctly abstains on absent objects.

Tokens: 0->0 | Latency: 494ms | Time: 2026-09-28 20:07:12

Test 2: Absent Object - Wrong Colour - PASS | Score: 1.00

Input Prompt:

```
Locate all the instances that matches the following description: yellow triangle.
```

AI Response:

```
Model ref: yellow triangle all the instances that matches the following: Distinct boxes returned: 0
(repeated 0x in raw output) Expected boxes: 0 Correctly returned no boxes.
```

Method	Score	Status
Localization	1.00	PASS
Detection	1.00	PASS
Format	1.00	PASS

Observations: localization: 1.00 | detection: 1.00 | format: 1.00 | absent-object case: clean | probe: Neither the colour nor the shape is present.

Recommendations: Correctly abstains on absent objects.

Tokens: 0->0 | Latency: 536ms | Time: 2026-09-28 20:07:13

Test 3: Empty Image - PASS | Score: 1.00

Input Prompt:

```
Locate all the instances that matches the following description: red circle.
```

AI Response:

```
Model ref: red circle. the instances that matches the following description: Distinct boxes
returned: 0 (repeated 0x in raw output) Expected boxes: 0 Correctly returned no boxes.
```

Method	Score	Status
Localization	1.00	PASS
Detection	1.00	PASS
Format	1.00	PASS

Observations: localization: 1.00 | detection: 1.00 | format: 1.00 | absent-object case: clean | probe: Blank white image. Any box is invented.

Recommendations: Correctly abstains on absent objects.

Tokens: 0->0 | Latency: 569ms | Time: 2026-09-28 20:07:13

Category: SPATIAL_REASONING

Test 1: Spatial Reference - Leftmost - PASS | Score: 0.79

Input Prompt:

```
Locate all the instances that matches the following description: leftmost purple square.
```

AI Response:

```
Model ref: leftmost purple the instances that matches the following: leftmost purple square.
Distinct boxes returned: 3 (repeated 13x in raw output) Expected boxes: 1 expected px=[60, 200,
160, 300] -> predicted px=[60, 200, 161, 301] IoU=0.980 (match) extra prediction px=[481, 200,
581, 301] -> false positive extra prediction px=[269, 200, 371, 301] -> false positive
```

Method	Score	Status
Localization	0.98	PASS
Detection	0.50	WARNING
Format	1.00	PASS

Observations: localization: 0.98 | detection: 0.50 | format: 1.00 | mean IoU 0.980 over 1 target(s) | 2 false positive(s) | output repeated 13x (token waste) | probe: Three identical squares; only position disambiguates. Expect this to be hard.

Recommendations: Count is wrong (missed targets or false positives). Review instance separation for this scene type.

Tokens: 0->0 | Latency: 4304ms | Time: 2026-09-28 20:07:18

Category: SCALE_ROBUSTNESS

Test 1: Small Object Localization - PASS | Score: 0.95

Input Prompt:

```
Locate all the instances that matches the following description: red circle.
```

AI Response:

```
Model ref: red circle. the instances Distinct boxes returned: 1 (repeated 1x in raw output) Expected
boxes: 1 expected px=[300, 220, 340, 260] -> predicted px=[299, 219, 342, 261] IoU=0.886
(match)
```

Method	Score	Status
Localization	0.89	PASS
Detection	1.00	PASS
Format	1.00	PASS

Observations: localization: 0.89 | detection: 1.00 | format: 1.00 | mean IoU 0.886 over 1 target(s) | probe: A 40x40 target in a 640x480 frame (~0.5% of area).

Recommendations: Accurate localization and count. No action needed.

Tokens: 0->0 | Latency: 412ms | Time: 2026-09-28 20:07:18

Test 2: Cluttered Scene, Unique Target - PASS | Score: 0.99

Input Prompt:

```
Locate all the instances that matches the following description: orange triangle.
```

AI Response:

```
Model ref: orangeate all the instances that matches the following Distinct boxes returned: 1
(repeated 1x in raw output) Expected boxes: 1 expected px=[420, 300, 540, 420] -> predicted
px=[419, 299, 541, 421] IoU=0.967 (match)
```

Method	Score	Status
Localization	0.97	PASS
Detection	1.00	PASS
Format	1.00	PASS

Observations: localization: 0.97 | detection: 1.00 | format: 1.00 | mean IoU 0.967 over 1 target(s) | probe: Three shapes of different colours; target is unique but not salient.

Recommendations: Accurate localization and count. No action needed.

Tokens: 0->0 | Latency: 455ms | Time: 2026-09-28 20:07:19